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OFFICIAL ACADEMIC STUDY HANDBOOK & COURSE MANUAL

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SYSTEM SOFTWARE

System Software: Compilers, Assemblers, Loaders & Linkers

From Lexical Analysis & Parsing to Machine Code Translation, Relocation,
Dynamic Linking & System Architecture

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CHAPTER 1

Assemblers: Two-Pass Architecture & Macro Processors

1. Theoretical Foundations & Scope

Assemblers translate human-readable mnemonic code into machine-executable binary. This chapter explores forward reference resolution, symbol tables (SYMTAB), opcode tables (OPTAB), two-pass assemblers, and macro substitution.

2. Core Conceptual Framework

- **Pass 1 vs Pass 2 Functions:** Pass 1 determines memory offsets and builds symbol tables; Pass 2 generates object machine code.
- **Forward Reference Handling:** Resolving label addresses defined after their first reference using backpatching or two passes.
- **Macro Processors:** Macro definition tables (DEFTAB), parameter tables, and recursive macro expansion.

Key Practical Insights & Examination Takeaways:

- ' Trace location counter (LOCCTR) advancement through sample assembly programs.
- ' Differentiate assembler directives (e.g., START, END, RESW) from executable machine instructions.
- ' Explain macro parameter substitution and conditional macro expansion.

CHAPTER 2

Loaders and Linkers: Static vs Dynamic Linking & Relocation

1. Theoretical Foundations & Scope

Executable programs require linking and loading before execution. We analyze absolute loaders, relocating loaders, linking loaders, static versus dynamic link libraries (DLLs), and address relocation mechanics.

2. Core Conceptual Framework

- **Static vs Dynamic Linking:** Embedding object modules directly into executables versus runtime resolution of shared libraries.
- **Relocation Techniques:** Modification records and relocation bit masks adapting absolute addresses to dynamic load addresses.
- **Object File Formats:** Headers, text records, modification records, and end records in standard binary object formats.

Key Practical Insights & Examination Takeaways:

- ' Trace linking loader passes resolving external references (EXTREF and EXTDEF).
- ' Explain how shared libraries save system RAM and disk storage.
- ' Diagnose unresolved external symbol errors during compilation pipelines.

CHAPTER 3

Compiler Phases: Lexical Analysis, Parsing & Code Generation

1. Theoretical Foundations & Scope

Compilers bridge high-level programming languages with low-level hardware architectures. This chapter surveys compiler frontends (lexing with DFAs, context-free grammars, LL/LR parsing) and backends (three-address code, optimization, register allocation).

2. Core Conceptual Framework

- **Lexical Analysis (Scanner):** Converting character streams into tokens using regular expressions and Deterministic Finite Automata (DFA).
- **Syntax Analysis (Parser):** Constructing parse trees using top-down recursive descent or bottom-up shift-reduce parsing.
- **Intermediate Representation & Optimization:** Generating three-address code and applying dead code elimination, constant folding, and loop invariant code motion.

Key Practical Insights & Examination Takeaways:

- ' Construct parse trees from context-free grammar production rules.
- ' Differentiate ambiguous grammars and resolve precedence and associativity.
- ' Explain register allocation graph coloring algorithms and machine code synthesis.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

